

UNIVERSITY OF BOSASO

Garowe, Puntland-Somalia



جامعة بوساصو

جروي ، بونتلاند - الصومال

FACULTY: SCIENCE AND TECHNOLOGY

DEPARTMENT: SOFTWARE DEVELOPMENT

COURSE: ARTIFICIAL INTELLIGENCE

ID:BIT/13231/2022

AI Governance: Securing High-Stakes Decision Support Systems

As AI increasingly shapes decisions in finance, healthcare, and security, the systems become targets for manipulation. Ensuring safety requires understanding vulnerabilities, maintaining effective human oversight, and building a resilient governance structure.

Key Vulnerabilities

- Adversarial inputs that trick AI models into incorrect outputs.
- Data poisoning, where training data is manipulated to alter system behavior.
- Model theft or reverse-engineering, allowing attackers to exploit system logic.
- Over-automation, where unchecked AI decisions create cascading failures.
- Weak access controls leading to insider misuse or unauthorized tampering.

Balancing Automation and Human Oversight

Automation boosts speed and consistency, but high-risk contexts require human judgment. Human operators should supervise decisions that affect safety, rights, or significant financial outcomes. AI should defer to humans when confidence is low, data is unusual, or consequences are severe.

Effective oversight includes:

- Clear authority for humans to review, override, and halt AI decisions.
- Transparent explanations enabling meaningful interpretation.
- Real-time monitoring for anomalies and decision drift.

Resilient Governance Framework

- Risk-based system classification to determine required safeguards.
- Mandatory security audits, red-teaming, and continuous model testing.
- Strong access control, tamper-evident logs, and protected data pipelines.
- Incident reporting rules and accountability for developers and deployers.
- Regular updates to reflect new threat models and evolving standards.

A resilient framework ensures that automation supports—not replaces—responsible human judgment. With proper governance, AI becomes a trusted tool for critical decision-making rather than a new source of systemic risk.